Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

B.A. in Data Science and Applied Mathematics — GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science · Probability for Data Science · Introduction to Urban Data Analytics · *In progress:* Introduction to Machine Learning · Efficient Algorithms & Intractable Problems

Technical Skills

Languages: Python, SQL, Java

ML & Statistics: scikit-learn, XGBoost, LightGBM, PyTorch, Cross-Validation, Feature Engineering

Data: pandas, NumPy, GeoPandas, Matplotlib, seaborn, PostgreSQL, ArcGIS Online, Jupyter

Engineering: FastAPI, REST APIs, AWS S3, pytest, Git, GitHub

Experience

The After Collective | Fung Fellow, Health + Innovation | Berkeley, CA **Aug 2026 – Present**

- Helped run **7** interviews with wildfire survivors and leaders (led **2**), finding insurance a shared struggle.
- Turned that into **3** AI-assisted concepts with teammates, took on a checklist tool for survivors, and gathered reactions.
- Took on testing the partner's recovery coach for gaps, and organize the team's documentation.

Guang Labs | Engineering Intern | Remote **May 2026 – Aug 2026**

- Built a service for an AI video startup that sorts videos and images into **6** content types, so the platform can handle each type differently.
- Added a confidence check that retries uncertain results on long videos, and stored each label with its confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that tells users when their video finishes processing or fails, instead of leaving them waiting on an idle screen.

IDX Exchange | Data Science Intern | Remote **Jun 2026 – Aug 2026**

- Contributed to a California home-price estimator trained on **71K** single-family sales; the final model's typical estimate was off by **8.6%** (R^2 of **0.88**).
- Traced an unusable first result to a data leak, fixed it, and brought accuracy back to a usable range.
- Measured accuracy by price tier and across **161** school districts, showing estimates were least reliable for luxury homes (**13%** error vs. **6.5%** mid-priced) and resort markets.

Projects

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Apify, Factory Droid* GitHub

- Co-designed a transfer planner for California community college students that shows which classes count toward their major using official ASSIST data, citing a source for every answer.
- Shaped the product idea, features, and vision, and gave feedback that improved the interface and features.
- Kept it grounded in real data: credits are matched by code against official records, and campus opportunities are scraped from real pages (Apify, run by a Droid agent), with source links.

SightLine | *Python, PyTorch, OpenCV* GitHub

- ML team member on a mobile app that turns a day of Meta Ray-Ban footage into a **60**-second recap video.

Movie Genre Classification | *Python, scikit-learn, SentenceTransformers* GitHub

- Built a genre predictor for **42K+** films; chose Macro F1 so rare genres counted equally, raising it from 0.48 to **0.66**.
- Found a simple model beat boosted trees while training about **12x** faster, a speed vs. accuracy tradeoff.

Leadership & Involvement

UC Berkeley CDSS | Dean's Cabinet Transfer Representative | Berkeley, CA **Sep 2026 – Present**

- Advise CDSS leadership on the student experience for CDSS majors.

UC Berkeley CDSS | Reader/Tutor, CDSS 198 | Berkeley, CA **Aug 2026 – Present**

- Give resume feedback in a careers seminar, and run exam reviews, worksheets, and office hours for transfer students.

UC Berkeley EECS | Course Staff (UCS1), CS 61A | Berkeley, CA **Jun 2026 – Aug 2026**

- Taught **4** weekly exam-prep sections, held office hours, and helped write midterm and final questions.